

Drive Test Report

Mumbai , India

Generated on July 11, 2026

Table of Contents

1. Introduction	3
2. Area Summary	3
3. Drive Summary	4
4. Coverage KPI Analysis	6
5. Mobility KPI Analysis	16
7. Service KPI Analysis	21

1. Introduction

Data includes 23285 samples from 3 sessions.

2. Area Summary

Drive route covers key operational areas identified based on spatial distribution of samples and session density.

Hotspots & Marked Locations

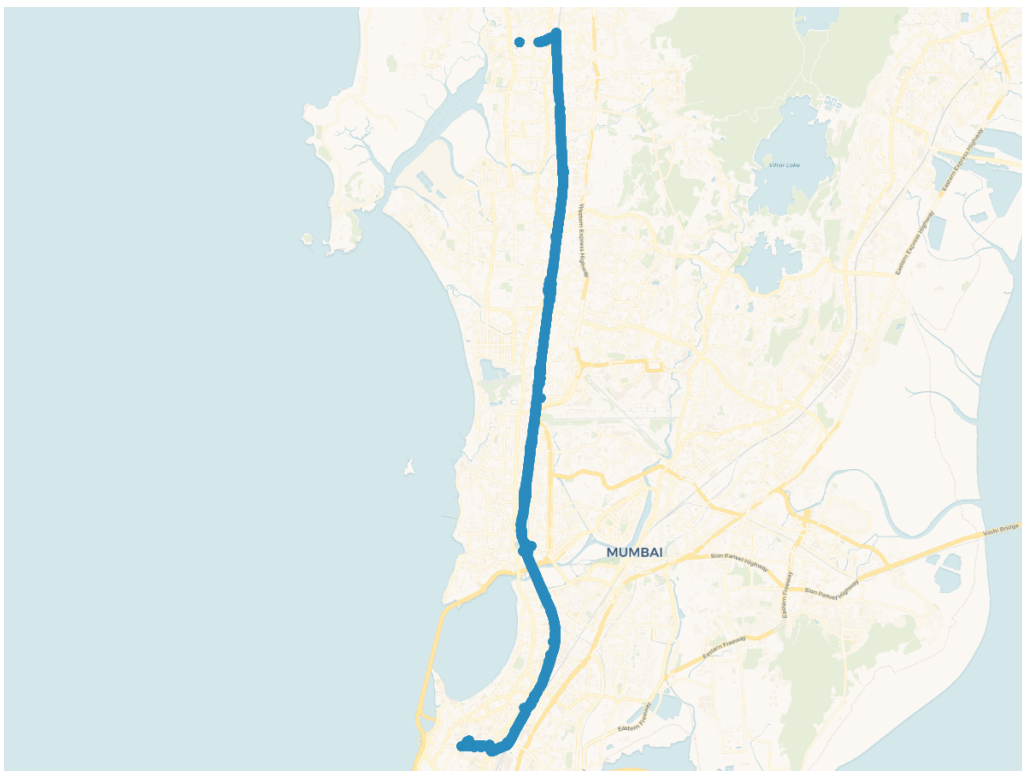
Hanuman Nagar, Balasheth Mandurkar Marg, Rai Pada

Crowded & High-Traffic Locations

Rai Pada

Major Areas Covered

The drive covered major areas including Hanuman Nagar (highway residential, 15811 samples), Balasheth Mandurkar Marg (highway residential, 835 samples), Rai Pada (highway residential, 612 samples), Kasturba Road (highway residential, 284 samples), Pandurang Budhkar Marg (highway secondary, 233 samples), Dadar West (highway primary, 171 samples).



3. Drive Summary

The drive test was conducted over 3 days from 2026-05-22 to 2026-06-05, covering 36.9 km with 23,285 samples collected across 3 sessions.

Metric	Value
Distance Covered	36.9 km
Total Samples	23,285
Total Sessions	3
Number of Days	3 Days

Session Details

Session	Date	Start	End	Duration	Distance
4530	22-05-2026	14:11	14:22	11 min	0.66 km
4714	04-06-2026	13:38	18:23	4 hours 45 min	20.57 km
4742	05-06-2026	07:14	08:10	55 min	15.62 km

Executive Summary

Objective

The objective of this drive test was to evaluate LTE radio coverage, radio quality, mobility performance and end-user service experience along the planned drive route. Measurements were collected using STracer and analysed against operator acceptance criteria.

Scope

- Coverage assessment using Band Distribution, RSRP, RSRQ and SINR.
- Mobility assessment using Serving Cell, PCI Distribution and Neighbor behaviour.
- Handover verification including Intra-frequency, Inter-frequency and Inter-RAT events where available.
- Service quality evaluation using Throughput, MOS, Latency, Jitter and Packet Loss.

Executive KPI Summary

KPI Group	Status	Remarks
Coverage	Good	Average RSRP -71.9 dBm; 94% of samples above -95 dBm.
Radio Quality - 4G (LTE Anchor - NSA)	Fair	[13,553 samples] Average SINR 19.5 dB; Average RSRQ -7.3 dB.
Radio Quality - 4G	Fair	[5,909 samples] Average SINR 18.7 dB; Average RSRQ -7.2 dB.
Radio Quality - 5G NSA	Fair	[3,823 samples] Average SINR 23.8 dB; Average RSRQ -10.3 dB.
Mobility	Good	129 unique serving PCIs (top PCI 41% of samples).
Handover	Good	685 band handover events; no major mobility interruption observed.

Key Engineering Observations

- Coverage classified as Good (94% of RSRP samples above -95 dBm, average -71.9 dBm).
- Radio quality (4G (LTE Anchor - NSA)) classified as Fair. Requires optimization.
- Radio quality (4G) classified as Fair. Requires optimization.
- Radio quality (5G NSA) classified as Fair. Requires optimization.

Overall Assessment

The network showed mixed RF performance with some KPI groups below the Good threshold. Targeted optimization is recommended for the KPI groups flagged Fair or Poor above.

4. Coverage KPI Analysis

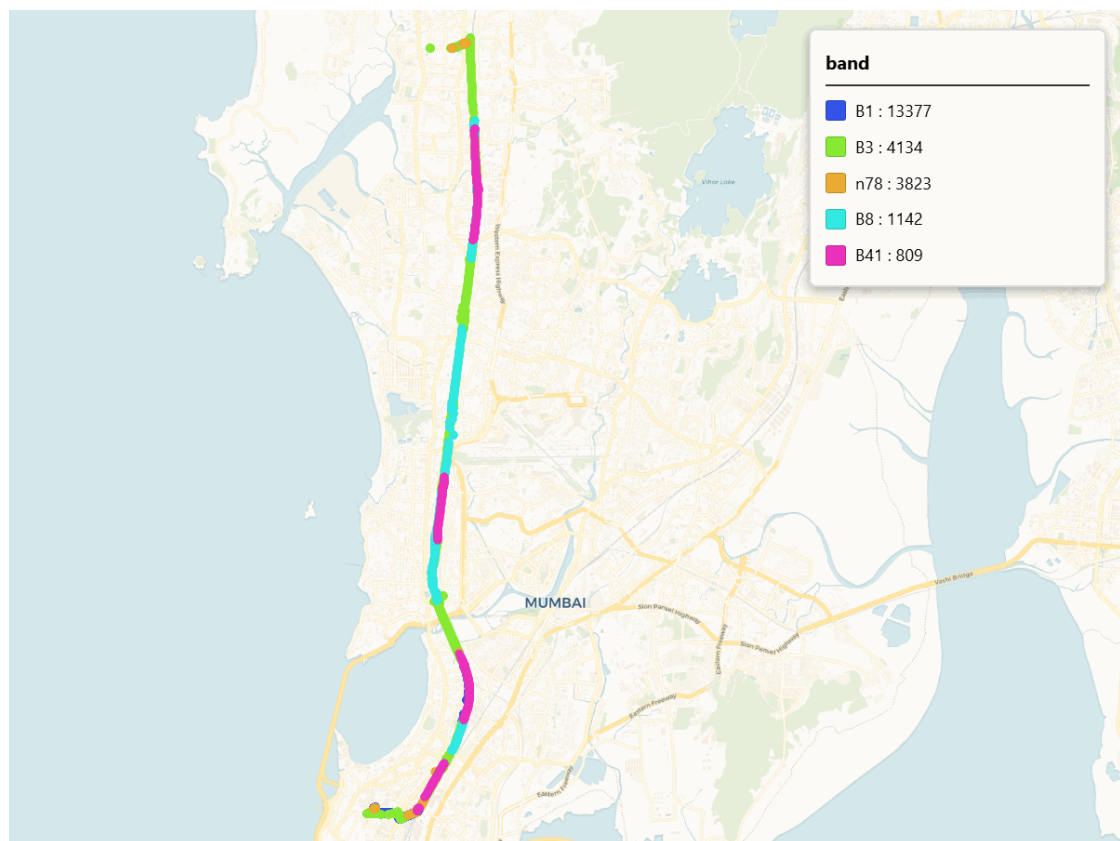
Coverage analysis is the first step in RF optimization. It evaluates whether the radio network provides continuous service with acceptable signal strength and quality across the planned drive route.

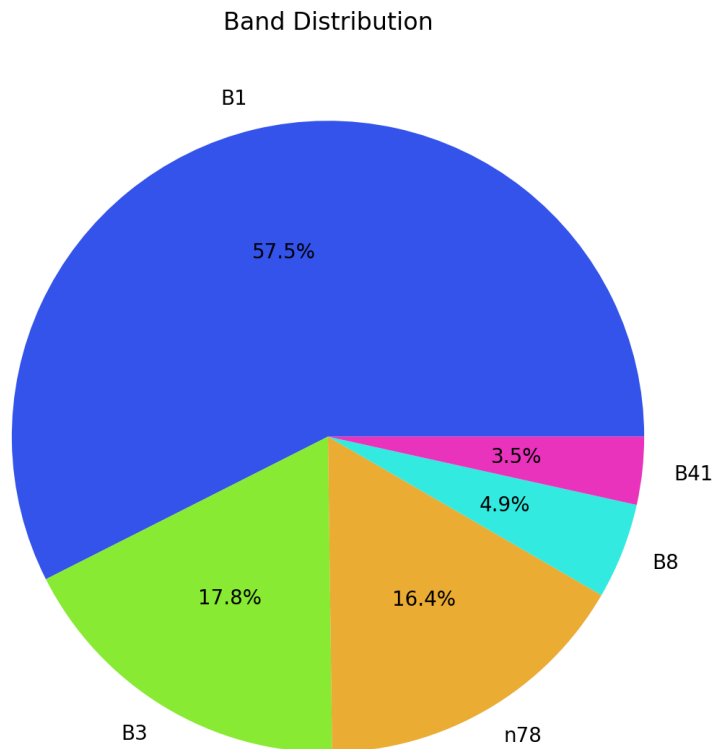
4.1 Band Distribution

Objective: verify that the UE camps on the planned band(s) throughout the drive route and identify unexpected band changes.

Band	Sample Count	% Samples
B1	13,377	57.5%
B3	4,134	17.8%
n78	3,823	16.4%
B8	1,142	4.9%
B41	809	3.5%

Observation: The majority of samples were served on Band B1 (57.5%), with additional coverage on Band B3 (17.8%), Band n78 (16.4%). No unexpected frequency changes were observed.





4.2 RSRP Analysis (Coverage)

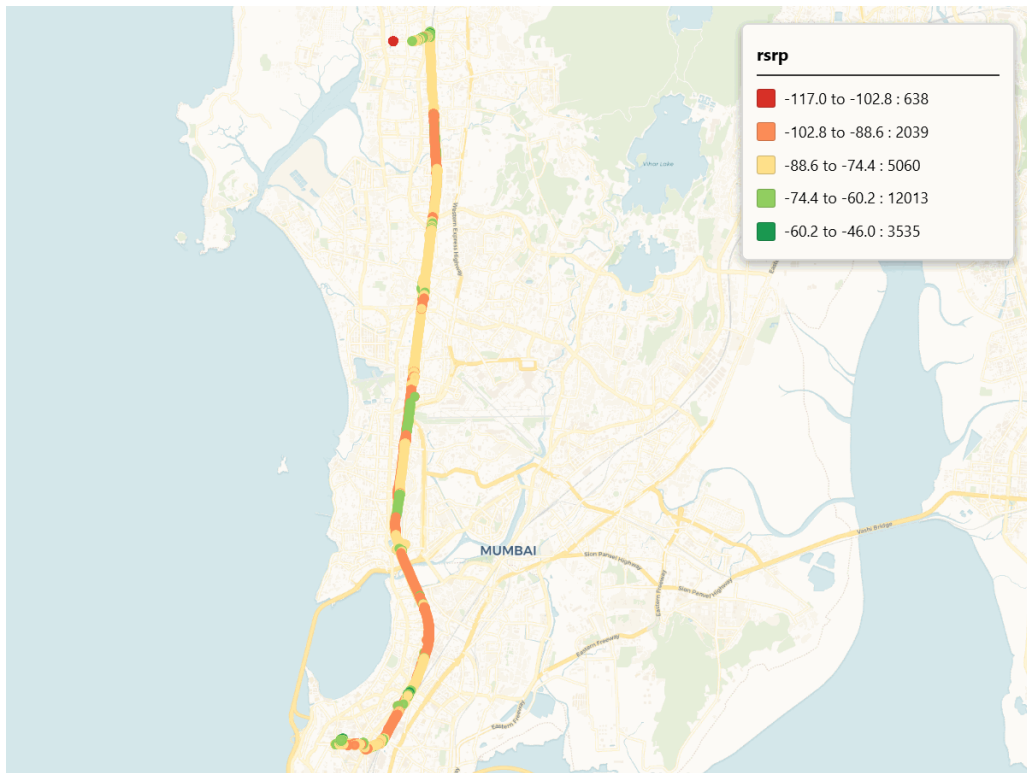
Definition: Reference Signal Received Power (RSRP) is the primary LTE coverage KPI and indicates the received signal strength from the serving cell.

Acceptance Criteria

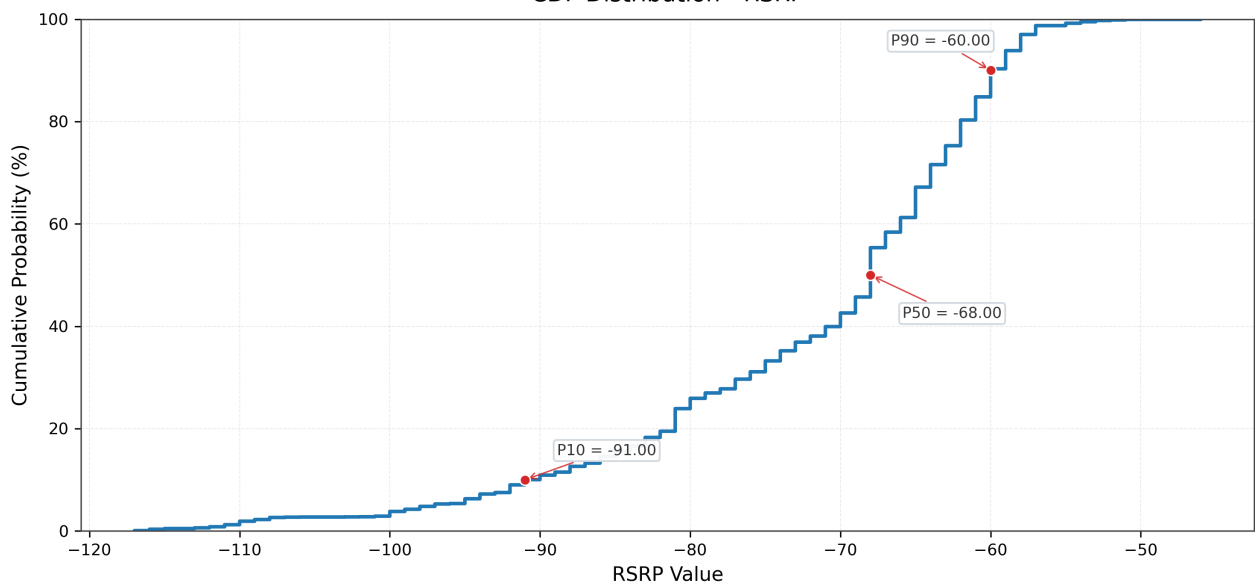
- Good: RSRP > -95 dBm for more than 90% of samples
- Fair: Average RSRP -95 to -105 dBm
- Poor: Average RSRP < -105 dBm, or coverage (% > -95 dBm) < 75%

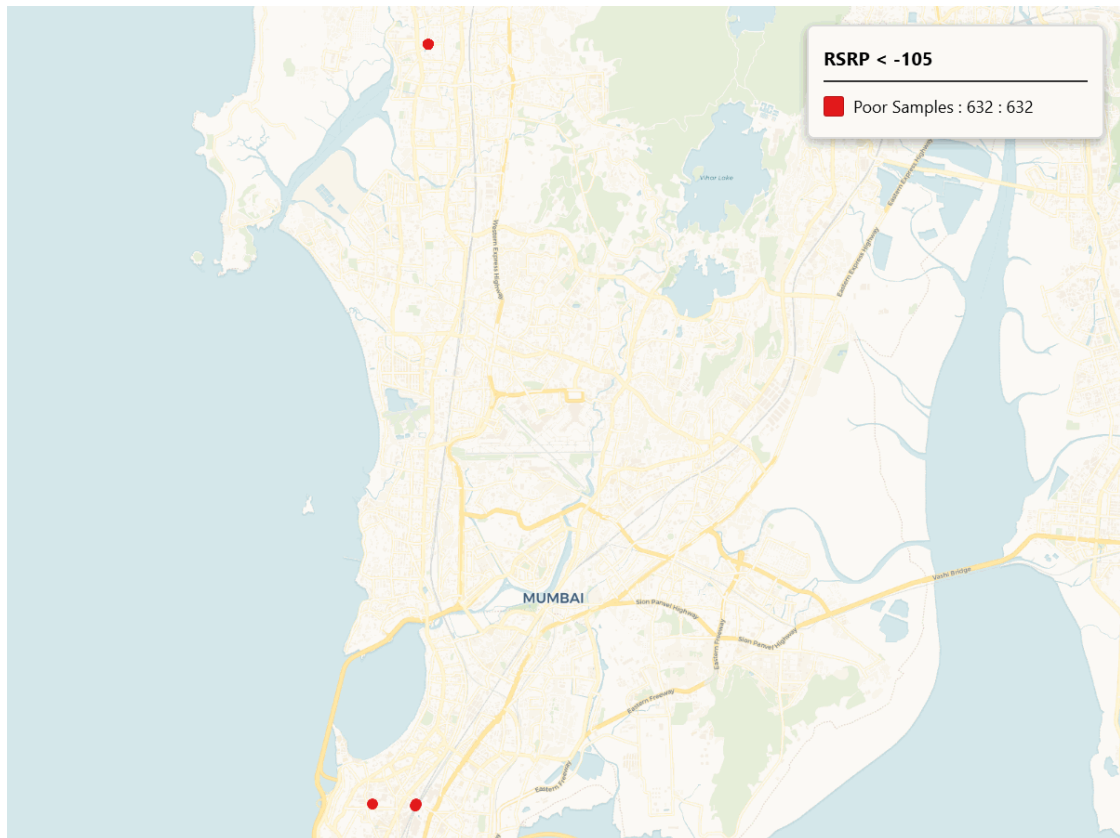
Metric	Sample Value
Average RSRP	-71.9 dBm
Best RSRP	-46.0 dBm
Worst RSRP	-117.0 dBm
Samples > -95 dBm	94%

Observation: Coverage classified as **Good**. Average RSRP -71.9 dBm; 94% of samples above -95 dBm.



CDF Distribution - RSRP





4.3 RSRQ Analysis

Definition: Reference Signal Received Quality (RSRQ) indicates the quality of the received reference signal and reflects both signal strength and cell loading. RSRQ is classified PER TECHNOLOGY, not blended, since its characteristic range differs by RAT.

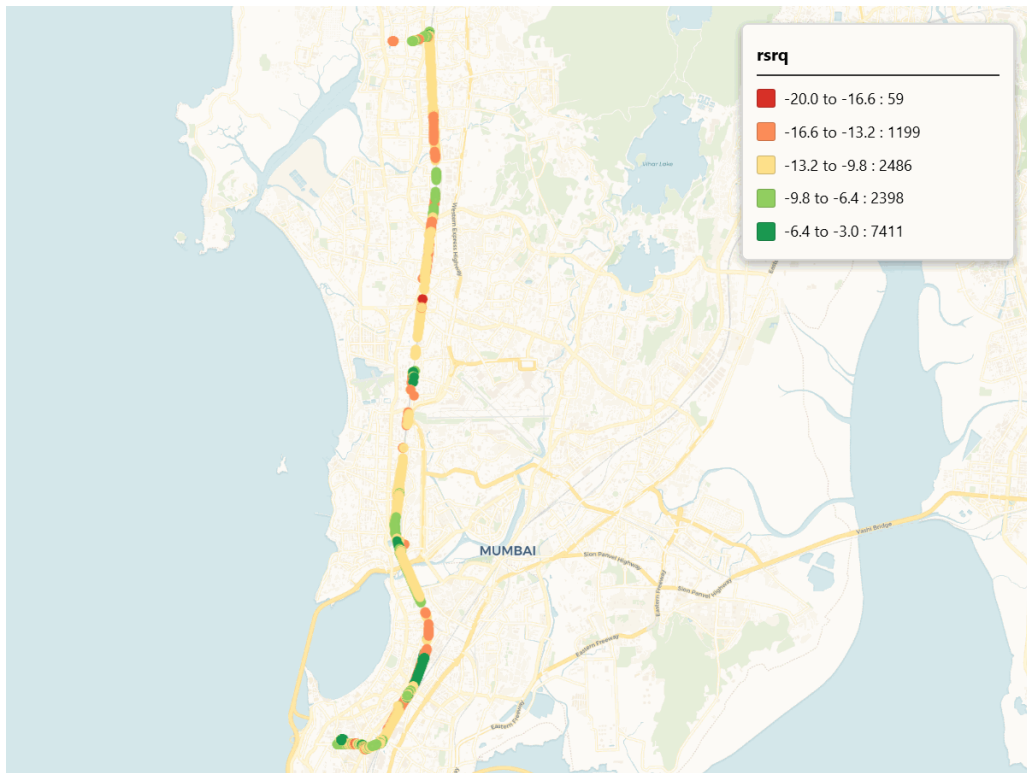
Acceptance Criteria

- Good: Average RSRQ > -10 dB
- Fair: Average RSRQ -10 to -15 dB
- Poor: Average RSRQ < -15 dB

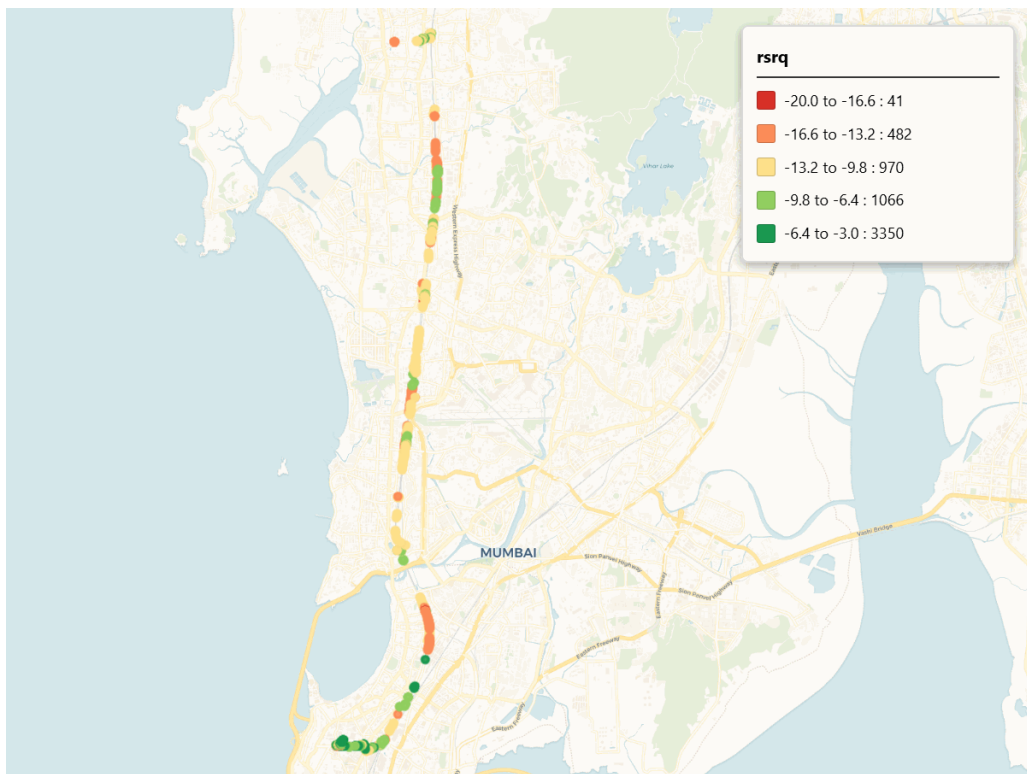
Technology	Average RSRQ	Worst RSRQ	Status
4G (LTE Anchor - NSA)	-7.3 dB	-20.0 dB	Good
4G	-7.2 dB	-20.0 dB	Good
5G NSA	-10.3 dB	-15.0 dB	Fair

Observation: worst-case RSRQ classification across technologies is **Fair**.

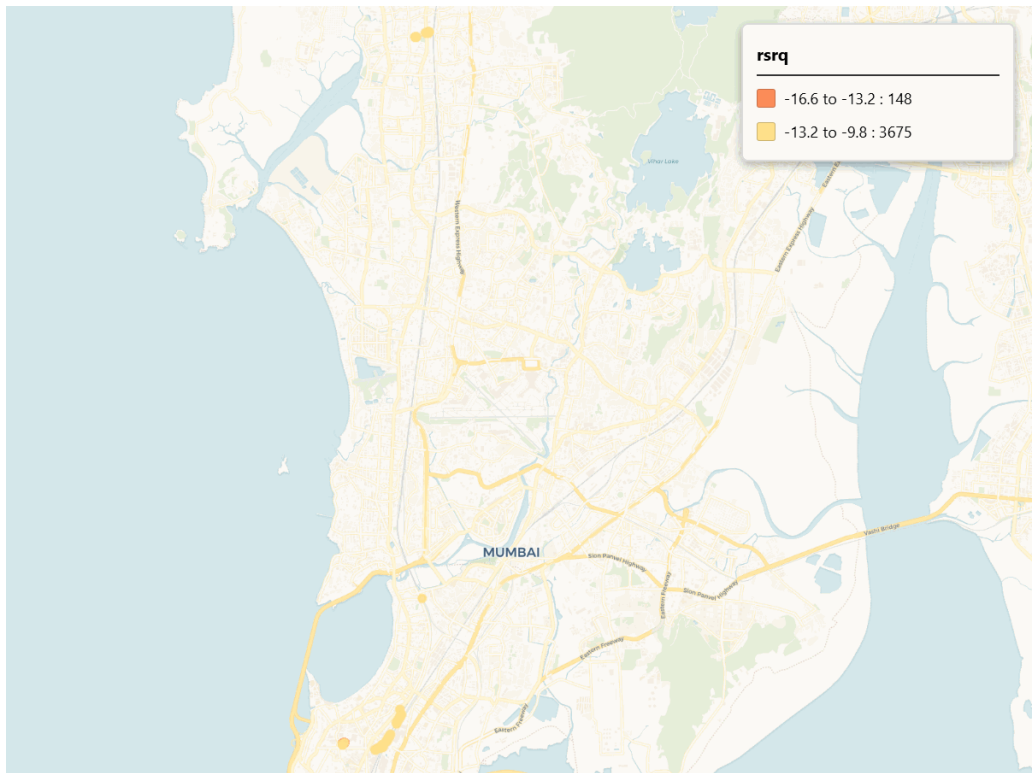
4G (LTE Anchor - NSA)



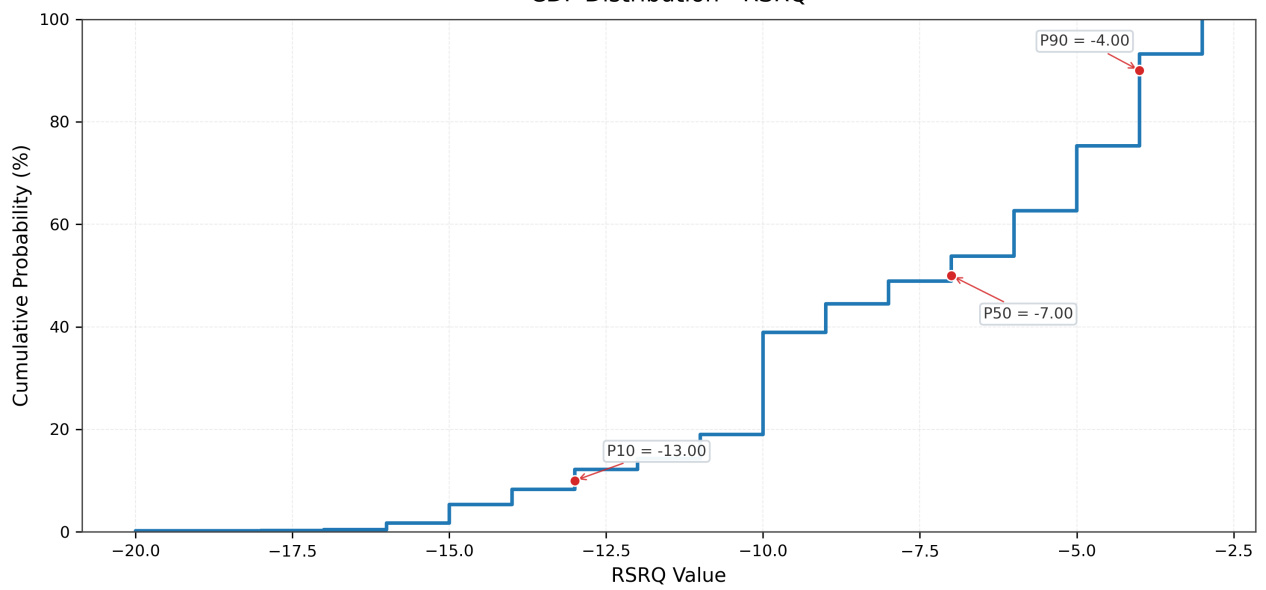
4G

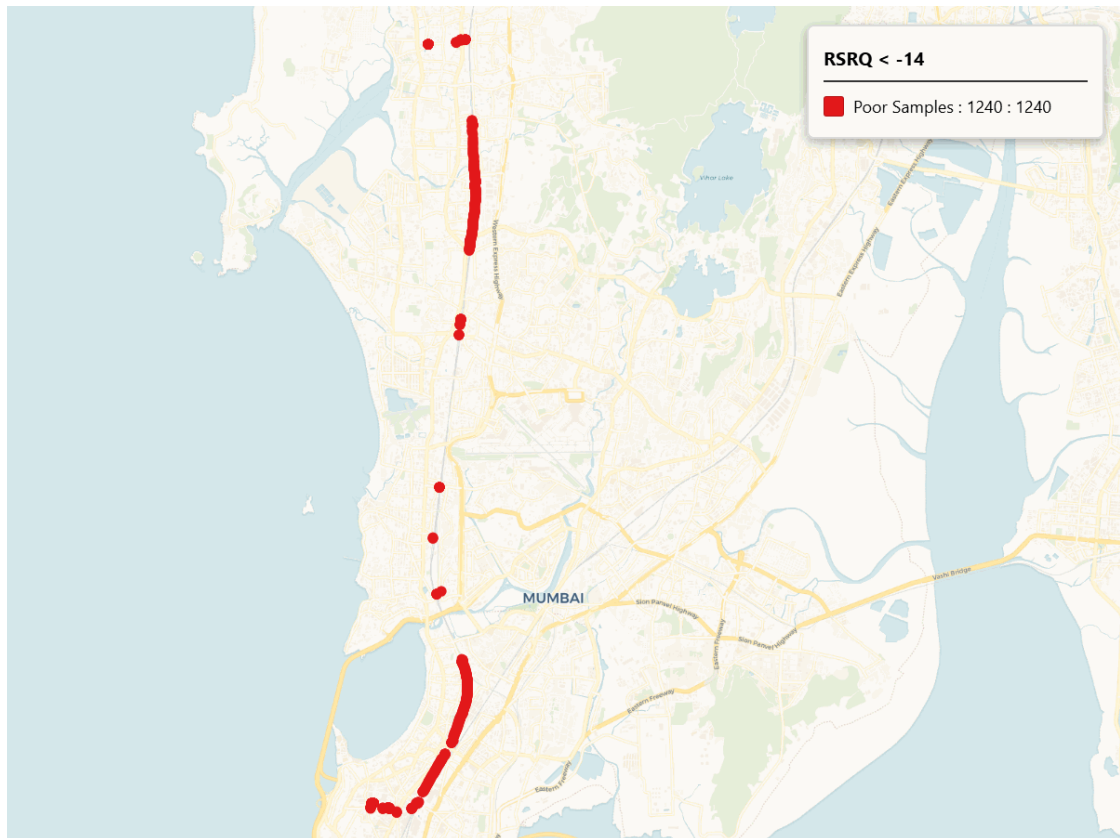


5G NSA



CDF Distribution - RSRQ





4.4 SINR Analysis

Definition: Signal-to-Interference-plus-Noise Ratio (SINR) represents radio link quality and interference conditions. SINR is classified PER TECHNOLOGY, not blended, since its characteristic range differs by RAT.

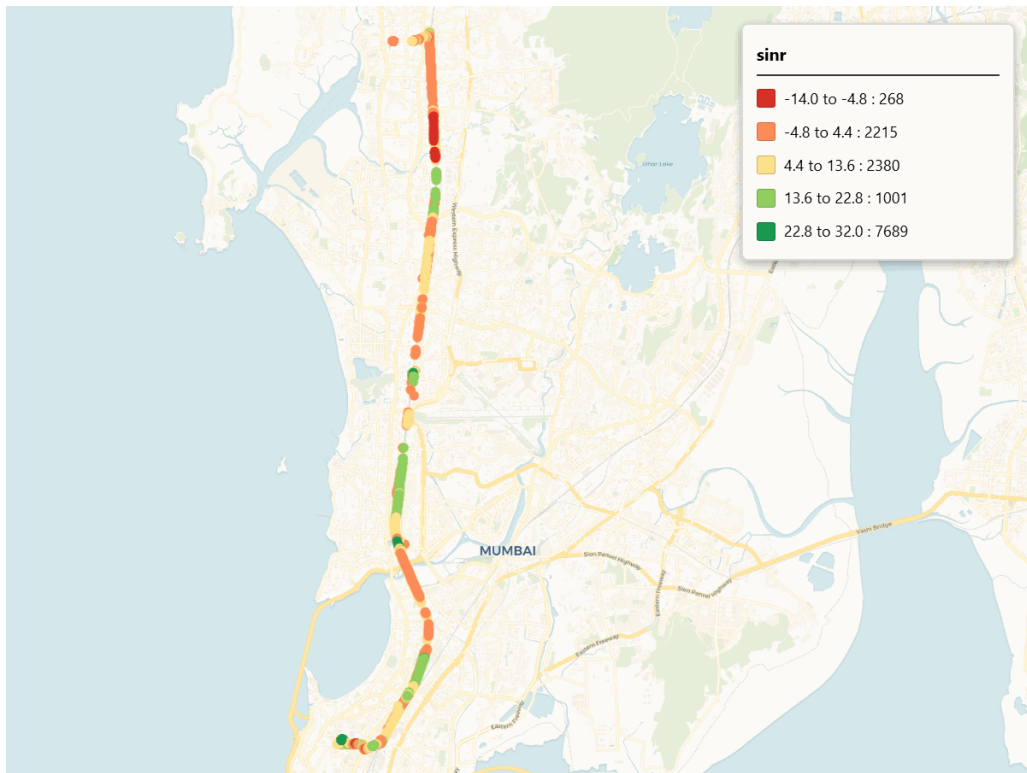
Acceptance Criteria

- Good: Average SINR > 20 dB
- Fair: Average SINR 13 to 20 dB
- Poor: Average SINR < 13 dB

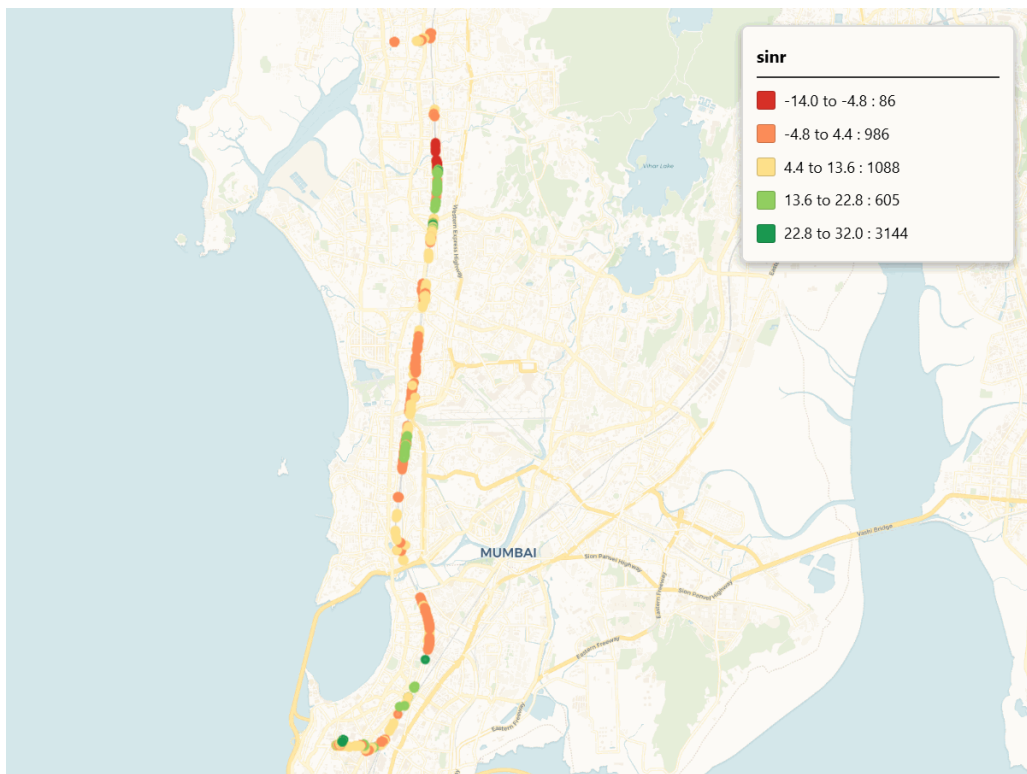
Technology	Average SINR	Worst SINR	Status
4G (LTE Anchor - NSA)	19.5 dB	-14.0 dB	Fair
4G	18.7 dB	-14.0 dB	Fair
5G NSA	23.8 dB	-4.0 dB	Good

Observation: worst-case SINR classification across technologies is **Fair**.

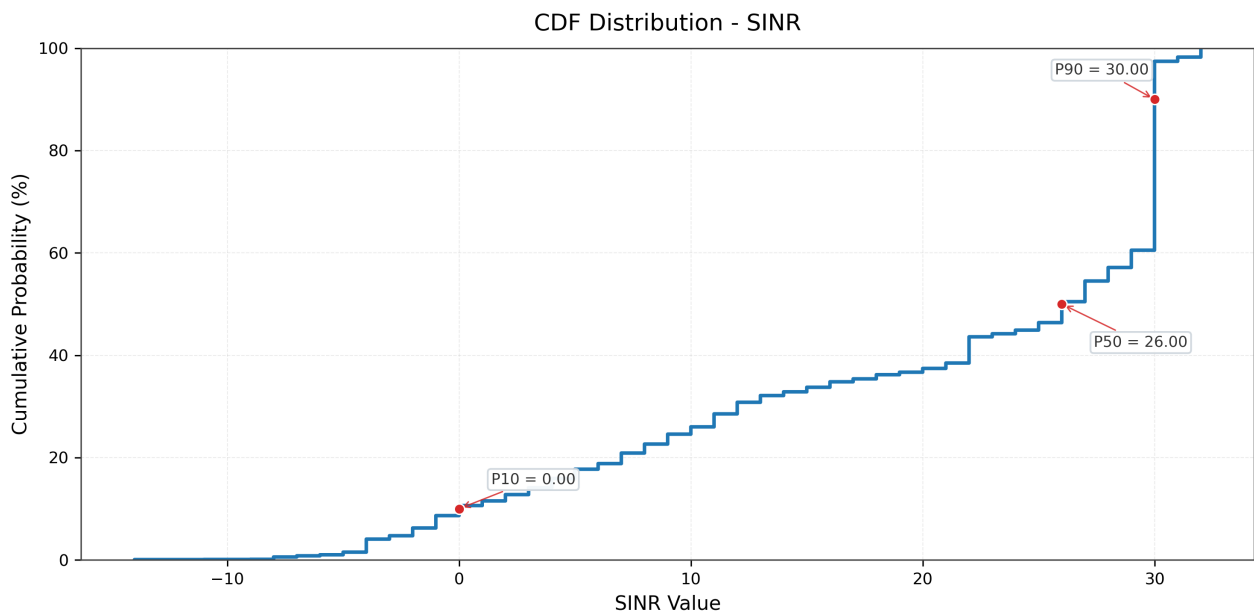
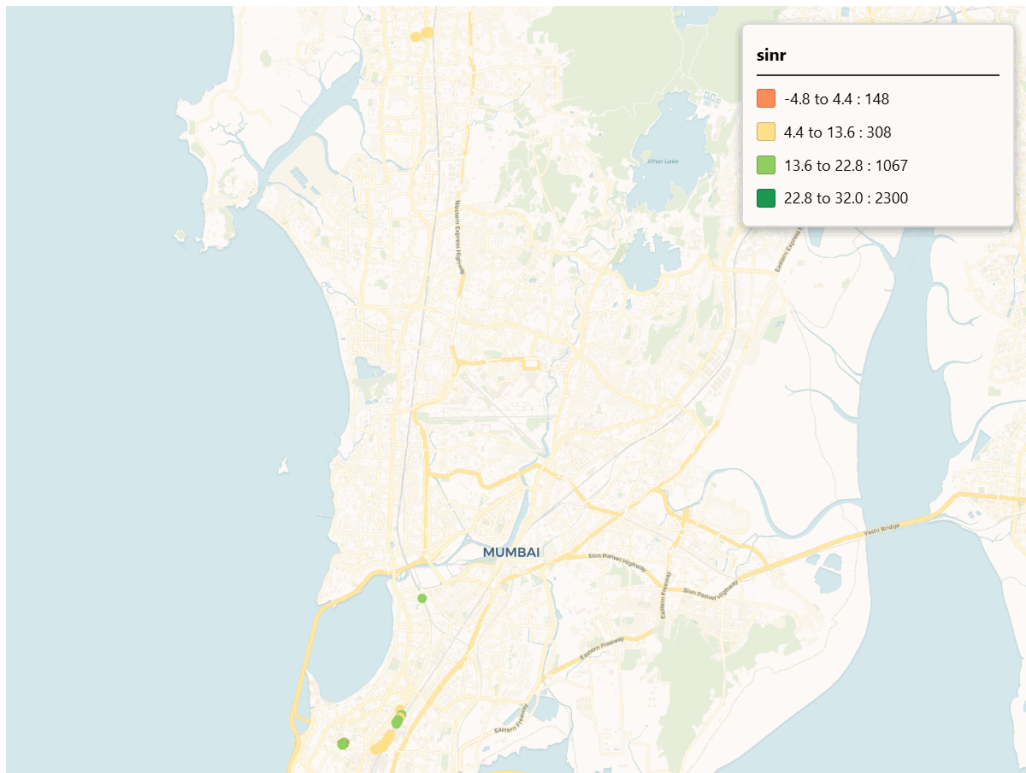
4G (LTE Anchor - NSA)



4G



5G NSA



4.5 Coverage KPI Summary

KPI	Acceptance	Observed	Status	% Within Acceptance	% Outside Acceptance
RSRP	> -95 dBm (90% of samples)	-71.9 dBm	PASS	94%	6%
RSRQ - 4G (LTE Anchor - NSA)	> -10 dB	-7.3 dB	PASS	72%	28%
RSRQ - 4G	> -10 dB	-7.2 dB	PASS	75%	25%
RSRQ - 5G NSA	> -10 dB	-10.3 dB	FAIL	0%	100%
SINR - 4G (LTE Anchor - NSA)	> 20 dB	19.5 dB	FAIL	59%	41%
SINR - 4G	> 20 dB	18.7 dB	FAIL	56%	44%
SINR - 5G NSA	> 20 dB	23.8 dB	PASS	87%	13%

Overall Coverage Assessment

The coverage KPIs indicate localized weak-signal or quality pockets. Targeted optimization is recommended for the specific KPI/technology combinations flagged Fair or Poor above.

5. Mobility KPI Analysis

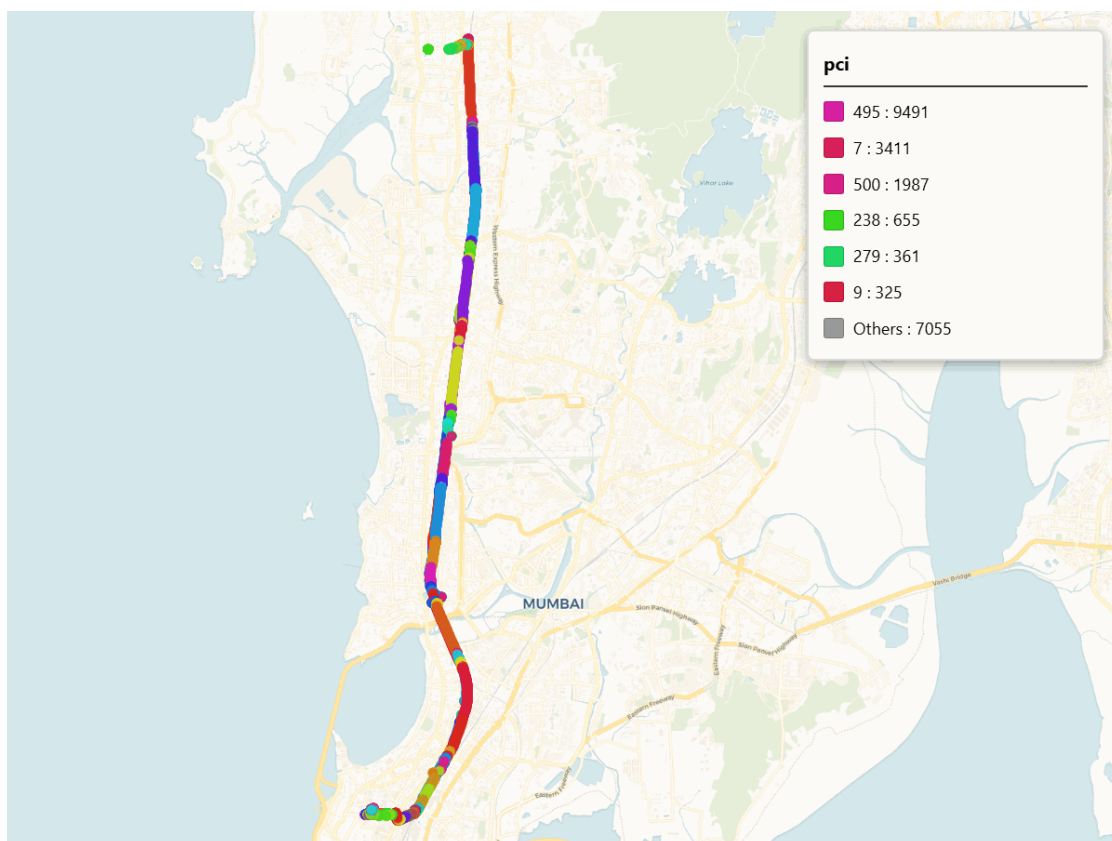
Mobility analysis evaluates whether the UE maintains service continuity while moving through the network. It focuses on serving cell behavior, PCI planning, neighbor relationships and cell dominance.

5.1 Serving Cell Distribution

Objective: verify that the UE remains connected to the expected serving cells across the planned route.

Metric	Observed Value
Samples with Serving PCI	23,285
Unique Serving PCIs	129
Dominant PCI	495
Dominant PCI Share	40.8%
Top 30 PCI Share	86.8%

Observation: Serving-cell transitions followed the planned footprint. 129 unique serving PCIs (top PCI 41% of samples).

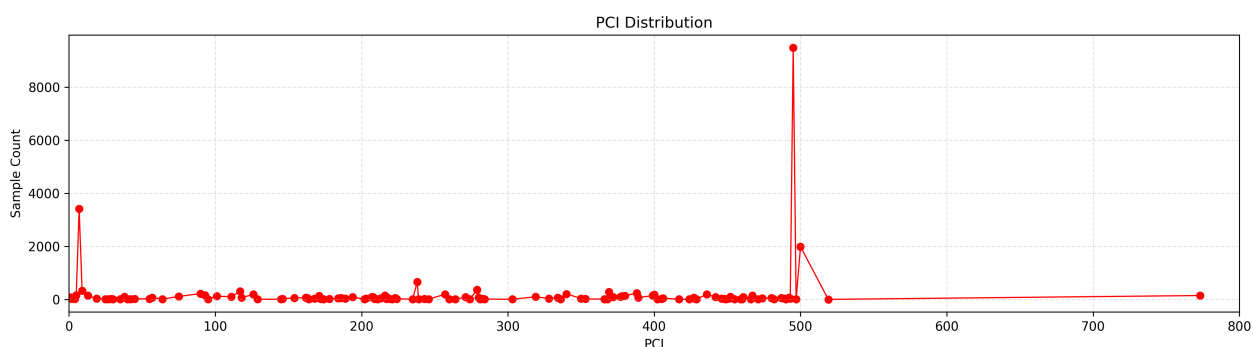


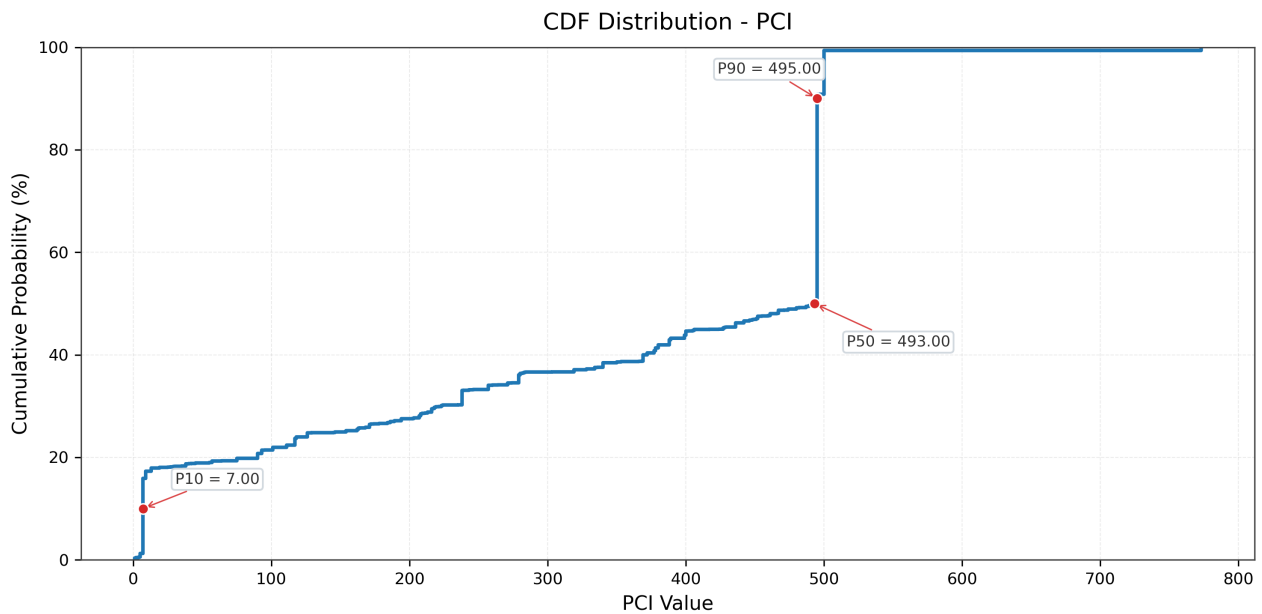
5.2 PCI Distribution

Objective: review how the serving PCI population is distributed along the drive route and identify whether mobility is concentrated on a small number of cells.

PCI	Sample Count	% Samples	CDF %
495	9,491	40.8%	40.8%
7	3,411	14.6%	55.4%
500	1,987	8.5%	63.9%
238	655	2.8%	66.8%
279	361	1.6%	68.3%
9	325	1.4%	69.7%
117	304	1.3%	71.0%
369	287	1.2%	72.2%
388	232	1.0%	73.2%
90	220	0.9%	74.2%
340	207	0.9%	75.1%
257	192	0.8%	75.9%
436	190	0.8%	76.7%
126	186	0.8%	77.5%
400	185	0.8%	78.3%

Observation: 129 unique serving PCIs were observed. The dominant PCI was 495 with 40.8% of samples. PCI allocation was generally balanced across the drive route.





5.3 Top PCI Analysis

PCI	Sample Count	% Samples	Observation
495	9,491	40.8%	Dominant serving PCI
7	3,411	14.6%	Healthy overlap
500	1,987	8.5%	Healthy overlap
238	655	2.8%	Normal utilization
279	361	1.6%	Normal utilization
9	325	1.4%	Normal utilization
117	304	1.3%	Normal utilization
369	287	1.2%	Normal utilization

Recommendation: review PCI reuse only if future expansion introduces additional neighboring sites.

5.4 Poor PCI Analysis (RSRP) (3 PCIs)

PCI	Poor Samples	Avg RSRP	Worst RSRP	Bands
238	612	-111.08	-117.0	B3
178	14	-107.0	-107.0	B41
427	6	-106.0	-106.0	B1

Observation: Weak PCI performance was localized and primarily associated with cell-edge conditions.

5.5 Poor PCI Analysis (RSRQ) (Top 15 of 32 PCIs)

PCI	Poor Samples	Avg RSRQ	Worst RSRQ	Bands
238	261	-15.41	-17.0	B3
90	173	-15.0	-15.0	B41
5	146	-15.0	-15.0	n78
436	144	-15.0	-15.0	B8
377	100	-15.0	-15.0	B41
442	87	-16.0	-16.0	B41
389	70	-16.0	-16.0	B3
495	25	-20.0	-20.0	B1
257	24	-15.5	-17.0	B1
7	20	-15.9	-16.0	B3,n78
19	18	-16.39	-17.0	B41
246	13	-15.46	-16.0	B3
168	12	-17.5	-18.0	B3
280	12	-15.5	-16.0	B1
283	12	-15.0	-15.0	B3

Observation: Low RSRQ values indicate loading or interference rather than pure coverage deficiency.

5.6 Neighbor Cell Analysis

Objective: verify that neighbor relations support smooth mobility.

Check	Status
Neighbor Logs	Observed
Unique Neighbor PCIs	9
Neighbor Consistency	Good

Observation: Dedicated neighbor logs are present in tbl_network_log_neighbour with 4,025 rows. Maintain periodic audit of neighbor relations after parameter changes.

5.7 Cell Dominance Analysis

Metric	Observed Value
Dominant Serving PCI	495
Serving PCI Share	40.8%
Dominant Neighbor PCI	7
Neighbor PCI Share	87.2%
Assessment	Serving dominance within acceptable range

Cell dominance was broadly consistent with the planned RF design. Overlap between adjacent sectors supported mobility without excessive overshooting dominance.

5.8 Mobility Summary

KPI	Acceptance	Observed	Status
Serving Cell Stability	Dominant PCI share < 80%	PCI 495 at 40.8% of samples	PASS
PCI Diversity	More than 1 serving PCI observed	129 unique serving PCIs	PASS
Neighbor Cell Availability	Neighbor log rows available	Available	PASS

Observation: Mobility performance was stable across the evaluated route. Serving-cell behaviour, PCI distribution and dedicated neighbor measurements support acceptable mobility conditions.

7. Service KPI Analysis

Service KPIs evaluate the end-user experience by measuring data performance, voice quality, application responsiveness and IP connectivity.

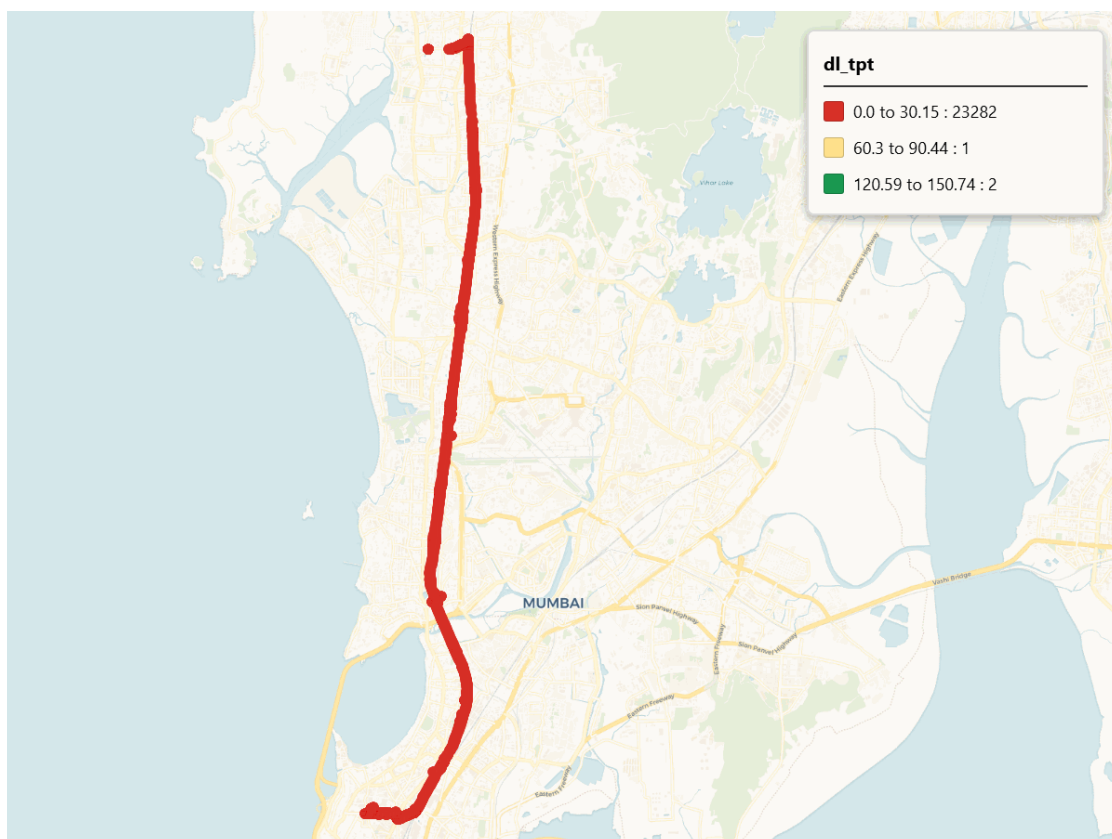
7.1 Downlink Throughput

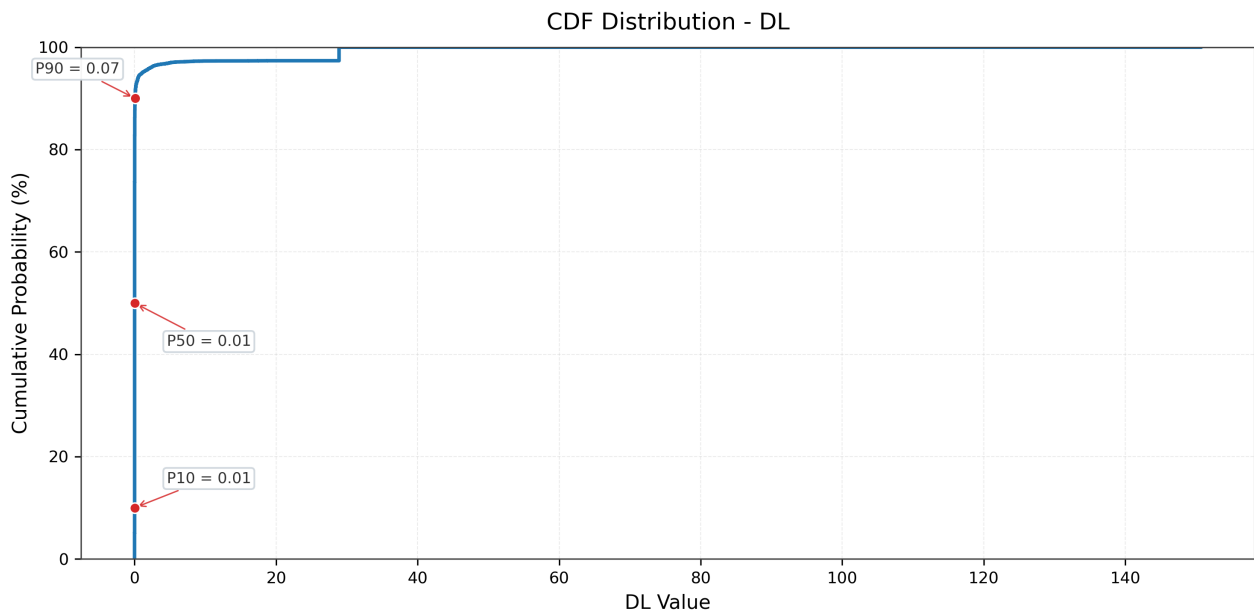
Objective: measure achievable download speed under live network conditions.

Metric	Sample Value
Average DL Throughput	0.9 Mbps
Peak DL Throughput	150.7 Mbps
Minimum DL Throughput	0.0 Mbps

Observation: Download throughput remained satisfactory across most of the route. Reduced throughput is typically associated with higher cell loading.

Recommendation: review scheduler utilization, carrier aggregation and cell loading.

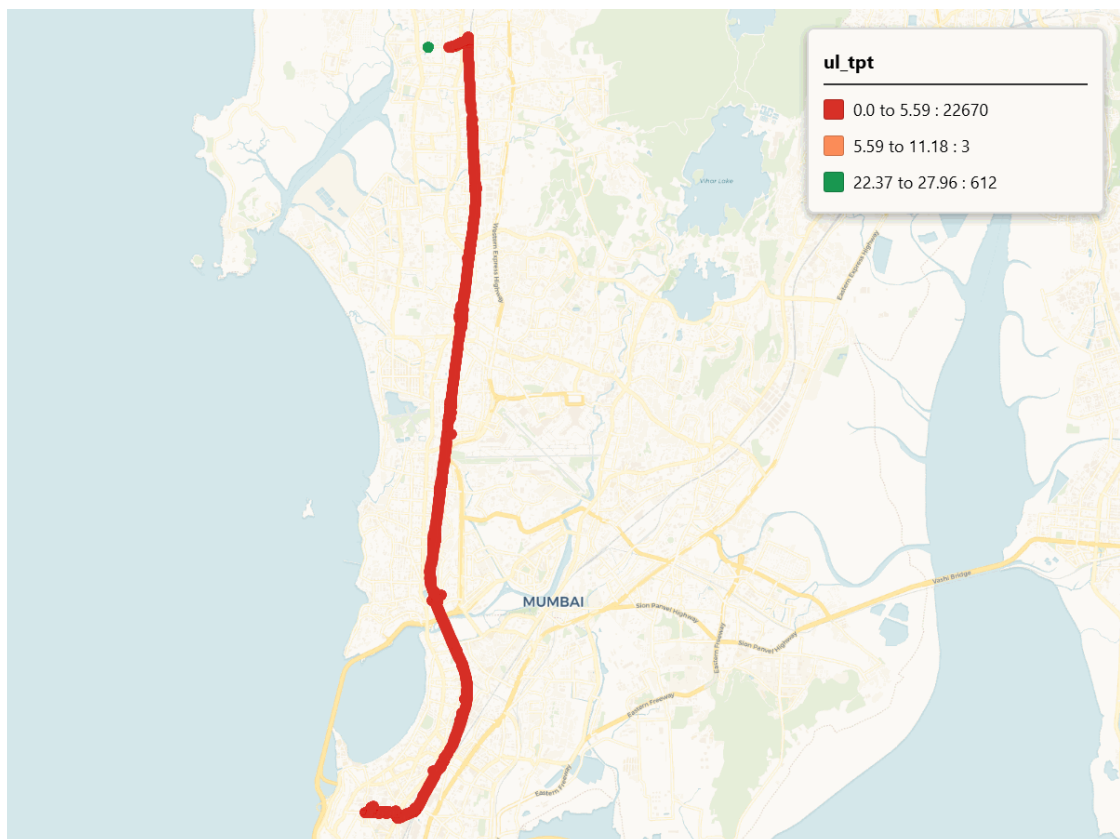


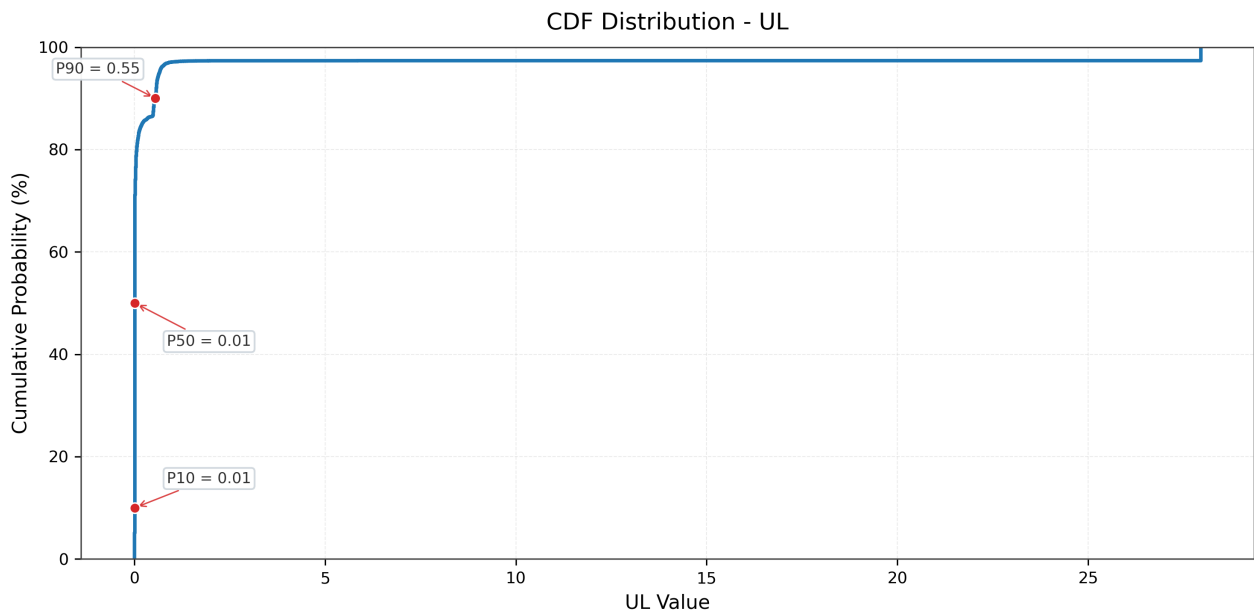


7.2 Uplink Throughput

Metric	Sample Value
Average UL Throughput	0.8 Mbps
Peak UL Throughput	28.0 Mbps
Minimum UL Throughput	0.0 Mbps

Observation: Uplink performance remained stable with isolated degradation in cell-edge regions.

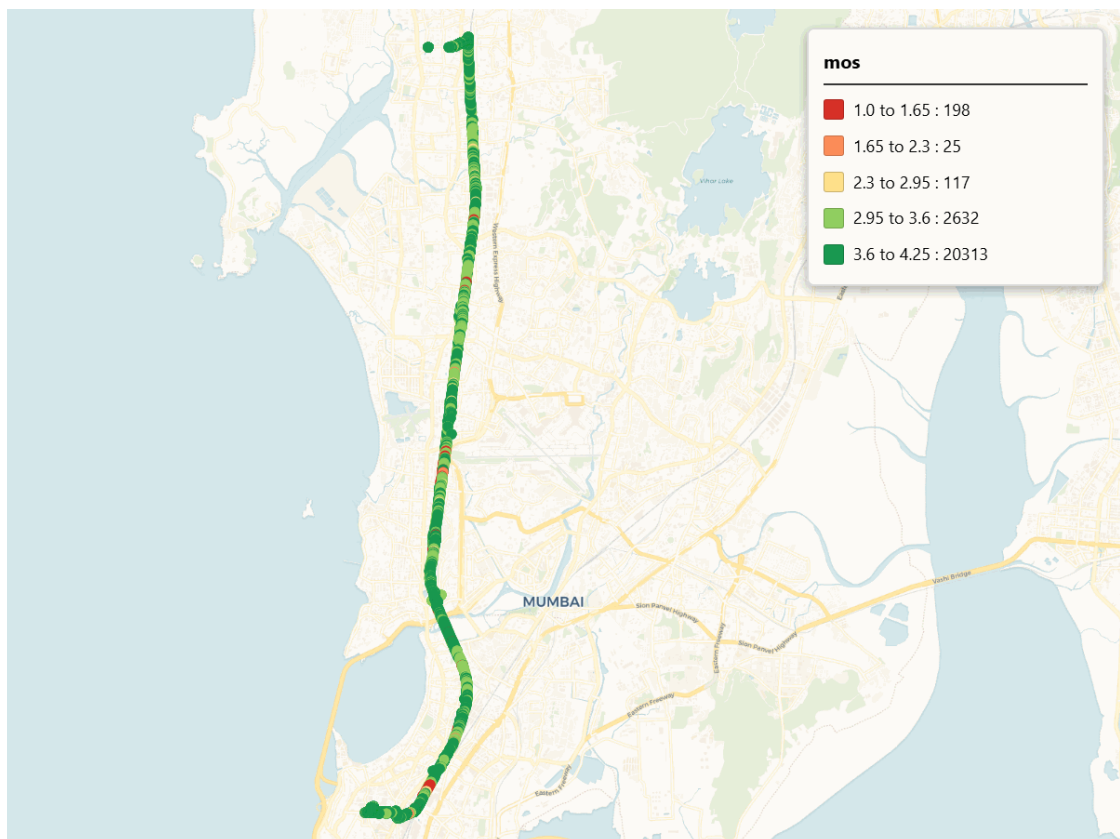


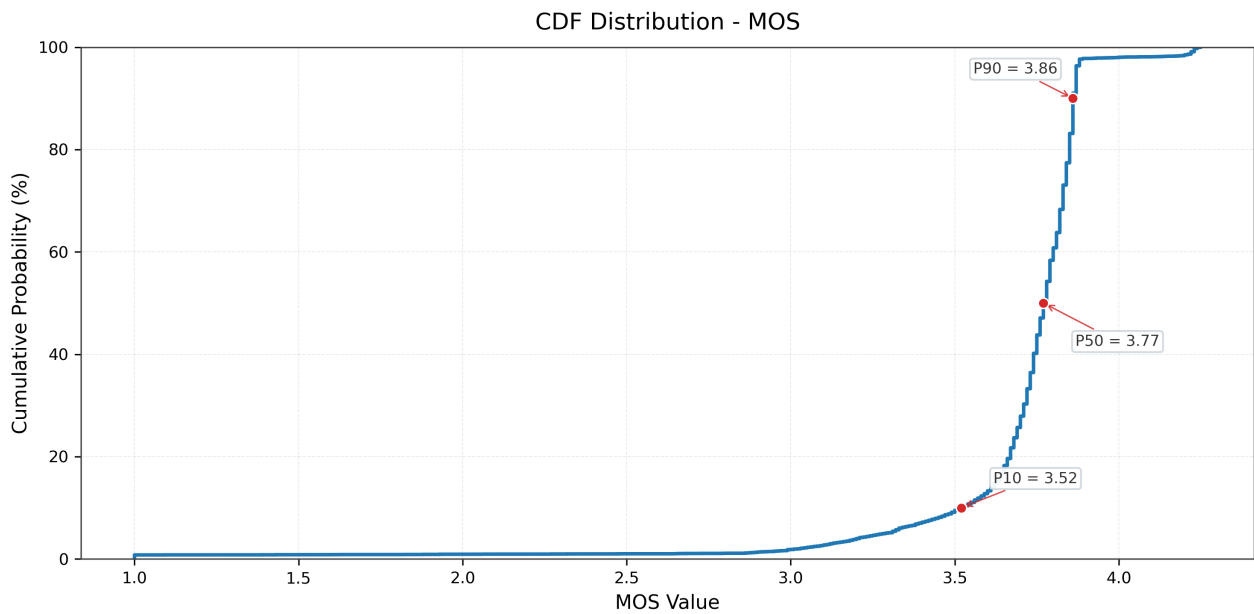


7.3 Voice Quality (MOS)

Metric	Sample Value
Average MOS	3.71
Minimum MOS	1.00

Observation: Voice quality remained acceptable with no perceptible degradation during successful calls.

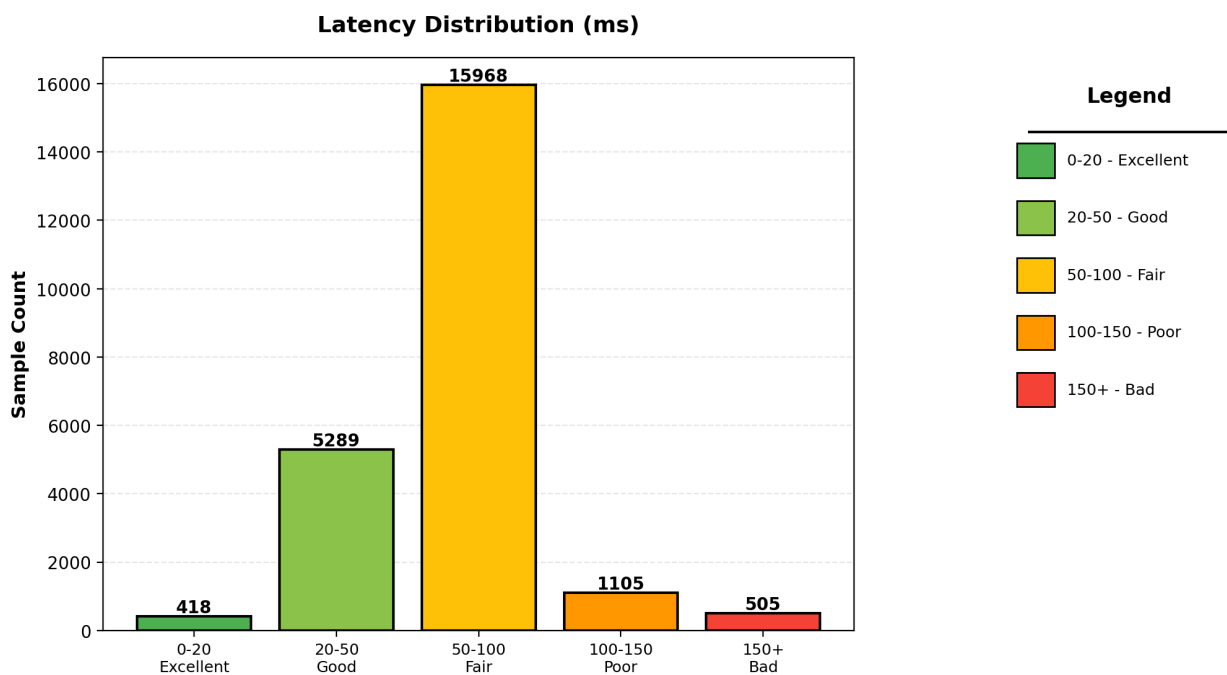




7.4 Latency Analysis

Metric	Sample Value
Average Latency	65.1 ms
Maximum Latency	1496.2 ms

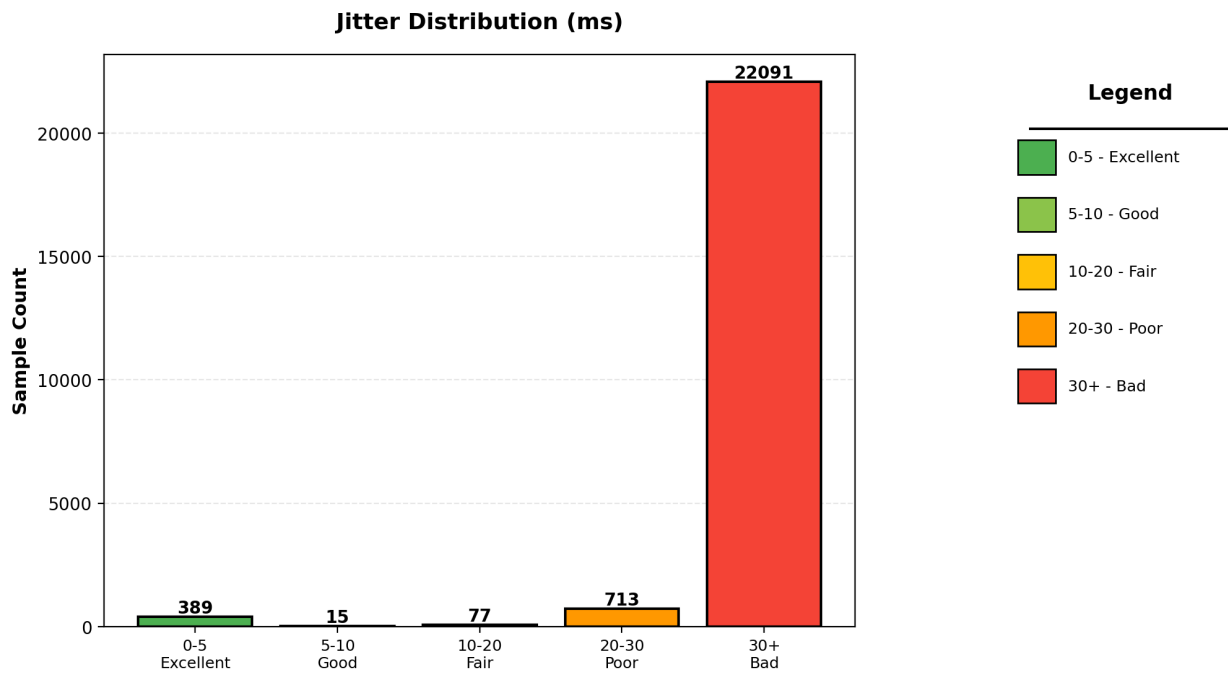
Observation: Latency values support satisfactory web browsing and real-time applications.



7.5 Jitter Analysis

Metric	Sample Value
Average Jitter	46.9 ms
Maximum Jitter	2058.8 ms

Observation: Jitter remained within acceptable limits for voice and video services.



7.6 Packet Loss

Metric	Sample Value
Average Packet Loss	0.22%

Observation: Negligible packet loss was recorded during the drive test.

7.7 Application Analytics

App Name	Sessions	Samples	Duration	RSRP Avg	RSRQ Avg	SINR Avg	DL Avg
Google Search	2	122	314:32:39	-71.3	-5.7	26.0	4.1 Mbps
Whatsapp Google Search	2	61	314:47:14	-100.5	-12.1	5.0	24.2 Mbps
Whatsapp	2	970	315:30:15	-71.5	-6.8	25.0	1.4 Mbps
Gmail Whatsapp	1	136	03:14:10	-63.6	-5.5	30.0	0.1 Mbps
Whatsapp Gmail	1	128	03:12:41	-67.0	-3.9	29.9	0.1 Mbps
Gmail	2	740	17:54:45	-66.9	-7.2	23.5	0.1 Mbps
Google Search Whatsapp	1	55	00:00:57	-67.2	-3.6	29.5	0.6 Mbps
Google Search Whatsapp Gmail	1	7	00:00:06	-66.3	-4.4	30.0	0.4 Mbps
Whatsapp Google Search Gmail	1	39	00:00:41	-62.5	-4.3	30.0	0.2 Mbps
Gmail Whatsapp Google Search	1	6	00:00:05	-63.0	-4.0	30.0	0.0 Mbps
YT	2	1500	14:27:01	-81.7	-10.3	7.6	0.3 Mbps
YT Whatsapp	1	101	00:02:13	-82.3	-9.1	9.9	0.3 Mbps
Whatsapp YT	1	20	00:00:20	-81.0	-9.0	11.0	0.3 Mbps
Google Search YT	1	7	00:00:06	-74.3	-7.0	5.0	0.8 Mbps

App Name	UL Avg	MOS Avg	Latency Avg	Jitter Avg	Loss Avg
Google Search	3.3 Mbps	3.85	48.9 ms	34.8 ms	0.0 %
Whatsapp Google Search	23.4 Mbps	3.92	42.0 ms	22.4 ms	0.0 %
Whatsapp	1.3 Mbps	3.74	64.8 ms	49.6 ms	0.15 %
Gmail Whatsapp	0.1 Mbps	3.7	64.0 ms	36.8 ms	0.07 %
Whatsapp Gmail	0.1 Mbps	3.7	64.0 ms	36.3 ms	0.0 %
Gmail	0.1 Mbps	3.74	59.7 ms	35.5 ms	0.09 %
Google Search Whatsapp	0.1 Mbps	3.78	55.7 ms	40.6 ms	0.0 %
Google Search Whatsapp Gmail	0.1 Mbps	3.6	74.1 ms	34.3 ms	1.43 %
Whatsapp Google Search Gmail	0.0 Mbps	3.38	95.8 ms	39.0 ms	1.03 %
Gmail Whatsapp Google Search	0.1 Mbps	3.34	99.8 ms	31.6 ms	0.0 %
YT	0.1 Mbps	3.62	80.7 ms	67.4 ms	0.43 %
YT Whatsapp	0.1 Mbps	3.47	93.3 ms	77.2 ms	0.0 %
Whatsapp YT	0.1 Mbps	3.66	67.6 ms	46.6 ms	0.5 %
Google Search YT	0.2 Mbps	3.74	60.2 ms	33.2 ms	0.0 %

Application	Observation
YouTube Streaming	Playback sensitive to throughput variation
WhatsApp	Stable messaging and media transfer

Application	Observation
Gmail	Mail sync and attachment loading remained usable
Web Browsing	Responsive

Observation: Application performance for YouTube Streaming, WhatsApp, Gmail reflected the overall network quality and remained generally stable during the drive.

7.8 Speed Test Summary

Test	Result
Download Speed	0.9 Mbps
Upload Speed	0.8 Mbps
Ping	65.1 ms

7.9 Service KPI Summary

KPI	Observed	Target	Status
DL Throughput	0.9 Mbps	Project Specific	PASS
UL Throughput	0.8 Mbps	Project Specific	PASS
MOS	3.71	≥ 3.5	PASS
Latency	65.1 ms	< 50 ms	FAIL
Packet Loss	0.22%	$< 1\%$	PASS

7.10 Engineering Conclusion

Overall service performance was satisfactory throughout the evaluated route. Data throughput supported common user applications, voice quality remained acceptable, and IP connectivity metrics stayed within acceptable limits. Localized degradation should be investigated through capacity optimization and scheduler analysis.